

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

```
In [2]: data = pd.read_csv(r'C:\Users\surya\OneDrive\Desktop\Project\DemographicData\data')
```

```
Out[2]:
```

	Country Name	Country Code	Birth rate	Internet users	Income Group
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income
...	...	...	...	...	...
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
191	South Africa	ZAF	20.850	46.5	Upper middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

195 rows × 5 columns

```
In [3]: x = data['Country Name'].head()
x
```

```
Out[3]: 0      Aruba
1      Afghanistan
2      Angola
3      Albania
4      United Arab Emirates
Name: Country Name, dtype: object
```

```
In [4]: y = data['Birth rate'].head()
y
```

```
Out[4]: 0      10.244
1      35.253
2      45.985
3      12.877
4      11.044
Name: Birth rate, dtype: float64
```

```
In [5]: p = data['Internet users'].head()
p
```

```
Out[5]: 0      78.9
1       5.9
2      19.1
3      57.2
4      88.0
Name: Internet users, dtype: float64
```

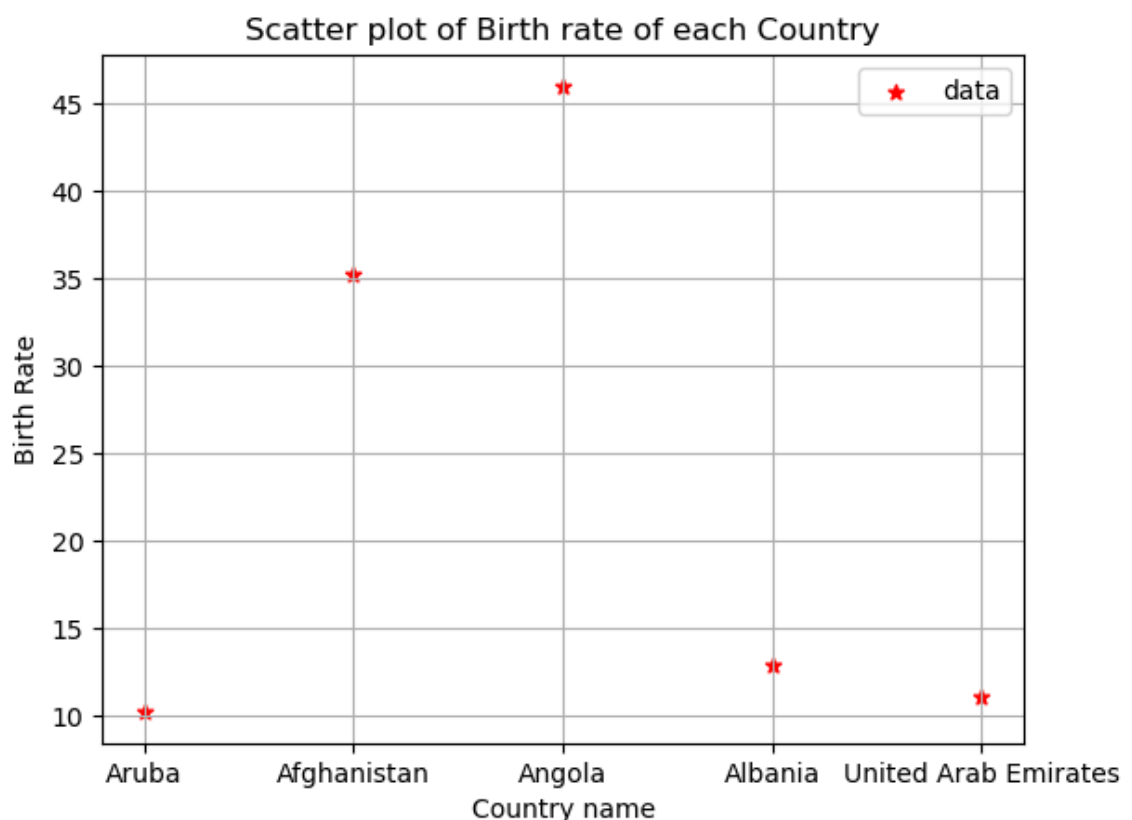
```
In [6]: i = data['Income Group'].head()  
i
```

```
Out[6]: 0          High income  
1          Low income  
2    Upper middle income  
3    Upper middle income  
4          High income  
Name: Income Group, dtype: object
```

Matplotlib

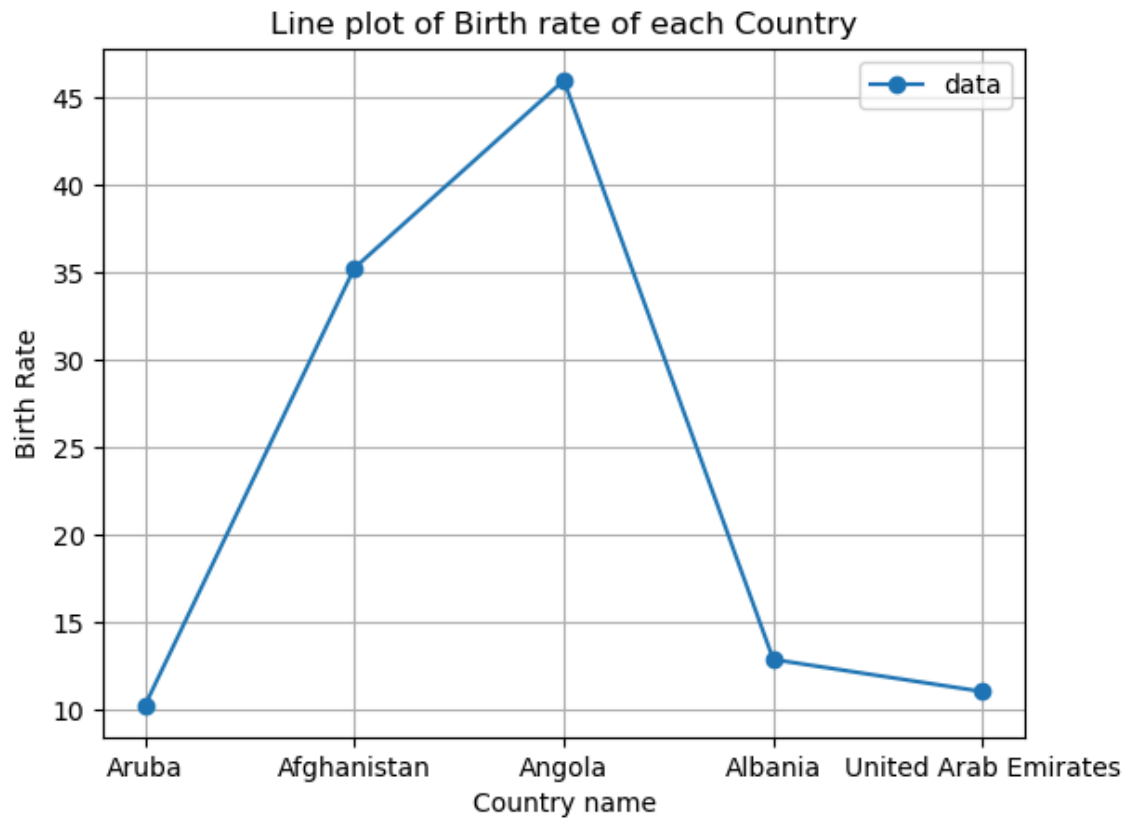
Scatter plot

```
In [7]: plt.scatter(x,y, marker = '*', color = 'red', label = 'data')  
plt.xlabel('Country name')  
plt.ylabel('Birth Rate')  
plt.title('Scatter plot of Birth rate of each Country')  
plt.grid(True)  
plt.legend()  
plt.show()
```



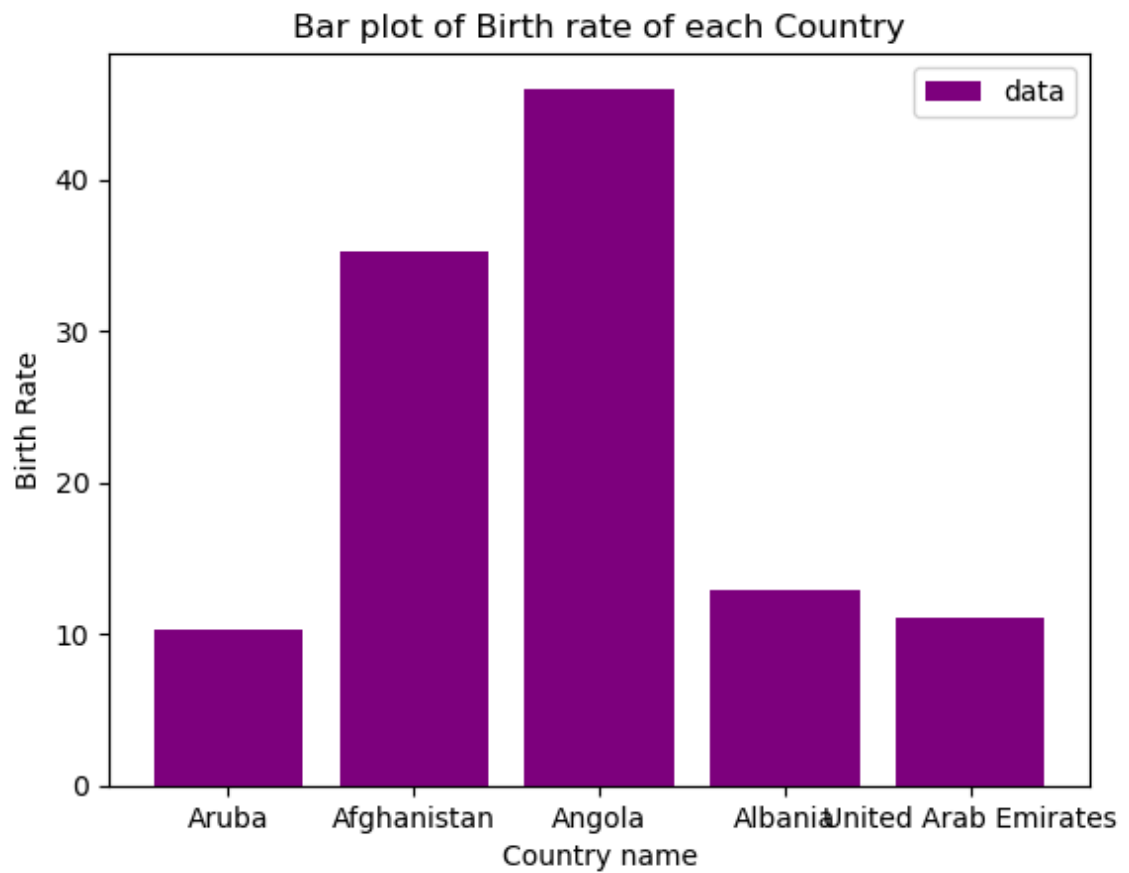
line plot

```
In [8]: plt.plot(x,y, marker = 'o', label = 'data')
plt.xlabel('Country name')
plt.ylabel('Birth Rate')
plt.title('Line plot of Birth rate of each Country')
plt.grid(True)
plt.legend()
plt.show()
```



Bar plot

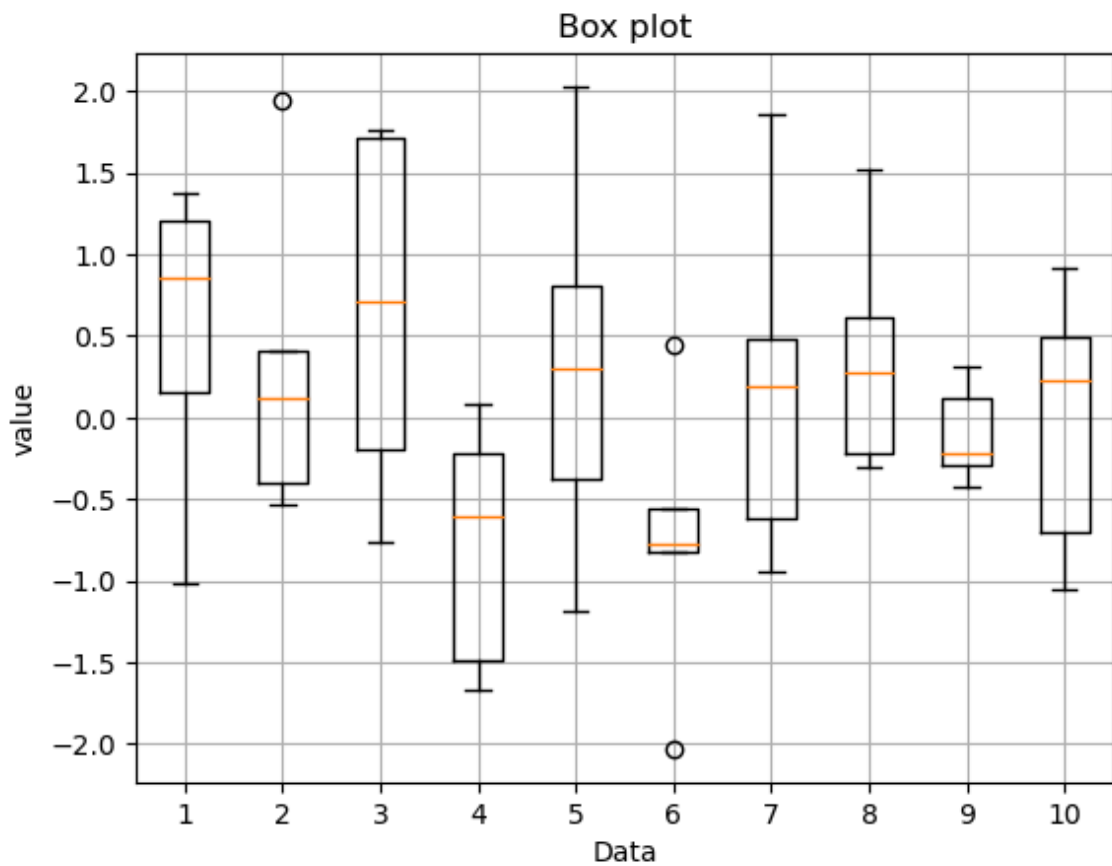
```
In [9]: plt.bar(x,y, color = 'Purple', label = 'data')
plt.xlabel('Country name')
plt.ylabel('Birth Rate')
plt.title('Bar plot of Birth rate of each Country')
plt.legend()
plt.show()
```



Box Plot

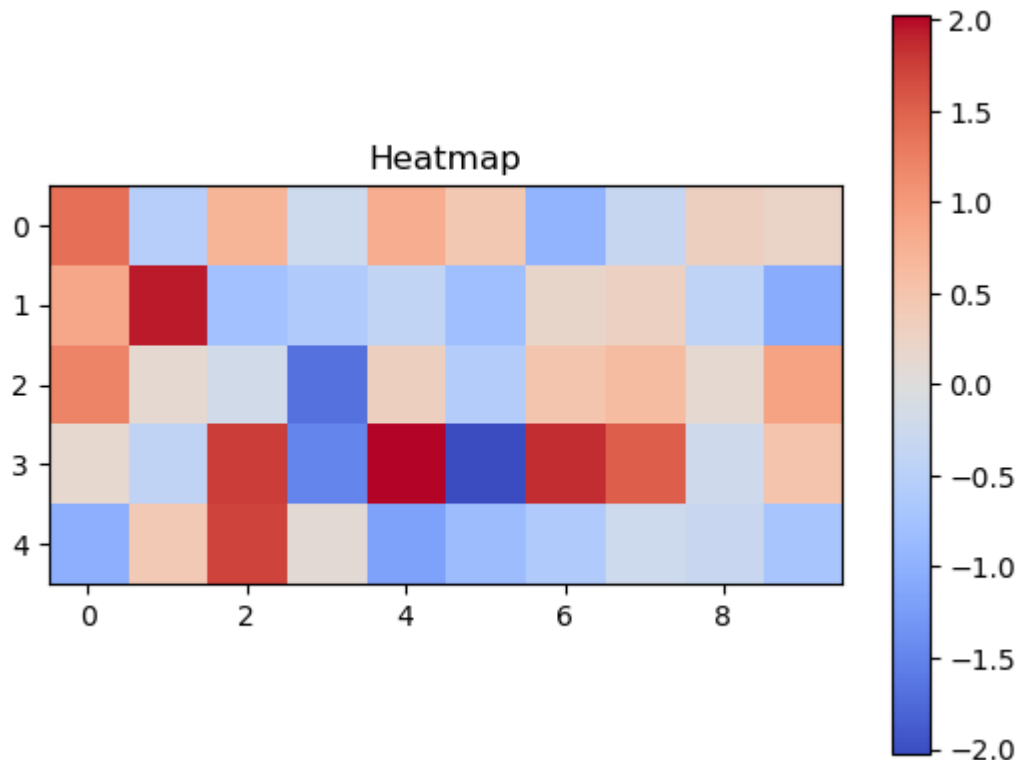
```
In [10]: data = np.random.randn(5,10)
```

```
plt.boxplot(data)
plt.xlabel('Data')
plt.ylabel('value')
plt.title('Box plot')
plt.grid(True)
plt.show()
```



Heatmap

```
In [12]: plt.imshow(data, cmap = 'coolwarm', interpolation = 'nearest')
plt.colorbar()
plt.title('Heatmap')
plt.show()
```

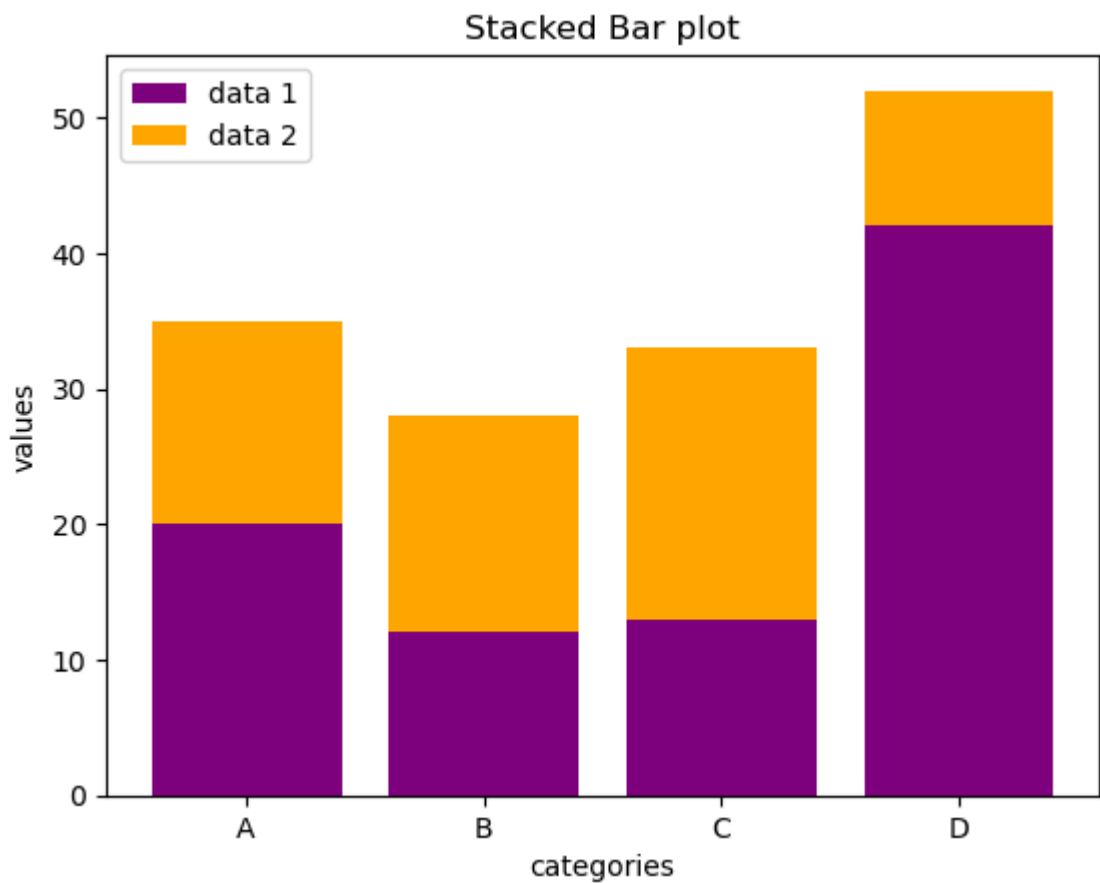


```
In [13]: data
```

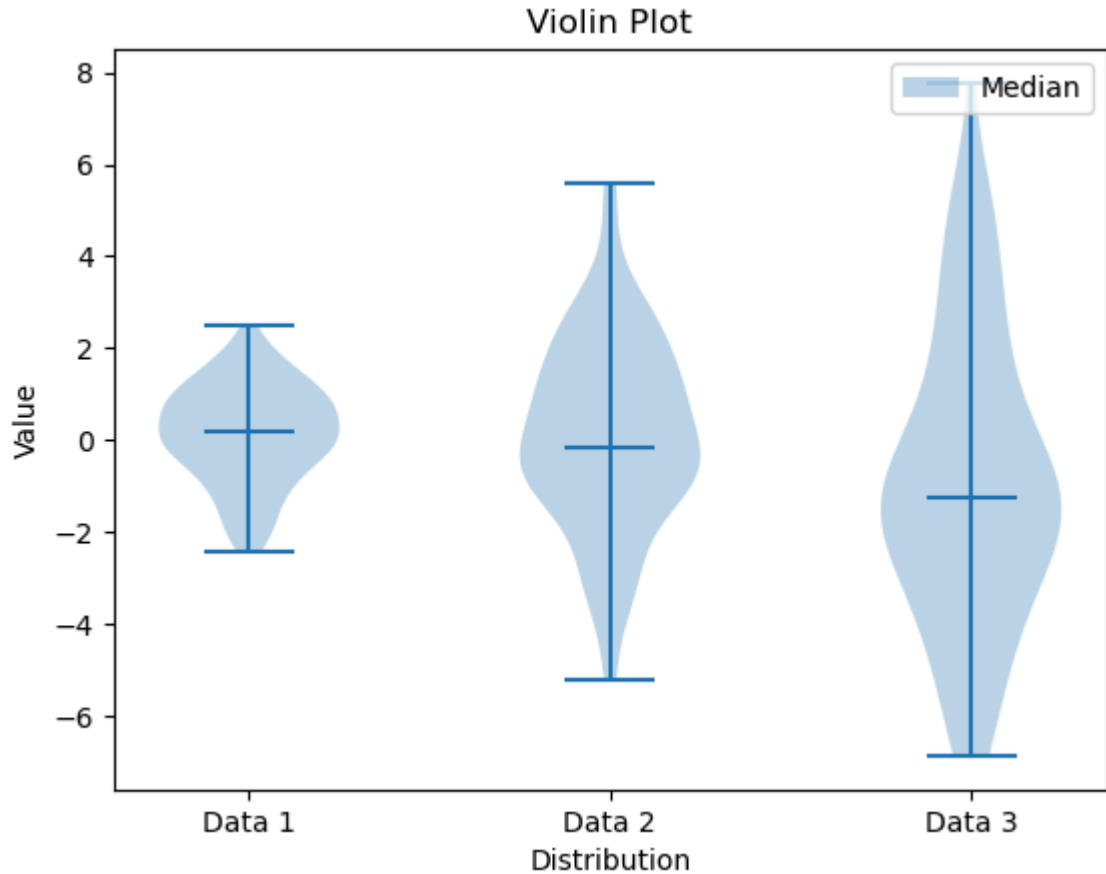
```
Out[13]: array([[ 1.37879434, -0.53195212,  0.70934644, -0.22513788,  0.80564583,
  0.44153329, -0.94484444, -0.30960169,  0.31469282,  0.22508193],
 [ 0.86115833,  1.94732742, -0.75815348, -0.61220522, -0.38233249,
 -0.77898567,  0.19110711,  0.27922097, -0.42415316, -1.05305725],
 [ 1.2064447 ,  0.11820855, -0.18977813, -1.67299077,  0.30362283,
 -0.55739886,  0.47646731,  0.61143112,  0.12055094,  0.91479734],
 [ 0.15356956, -0.40342297,  1.76795584, -1.48627911,  2.02904487,
 -2.03148609,  1.85457343,  1.52209779, -0.21553872,  0.49474207],
 [-1.01149014,  0.41200987,  1.71421959,  0.07874011, -1.1875515 ,
 -0.82973727, -0.6153567 , -0.22360005, -0.28820908, -0.69709876]])
```

```
In [14]: #scatter bar plot
```

```
In [19]: categories = ['A', 'B', 'C', 'D']  
values1 =[20,12,13,42]  
values2 = [15,16,20,10]  
  
plt.bar(categories, values1, color = 'Purple', label = 'data 1')  
plt.bar(categories, values2, color = 'orange', bottom = values1, label = 'd  
plt.xlabel('categories')  
plt.ylabel('values')  
plt.title('Stacked Bar plot')  
plt.legend()  
plt.show()
```



```
In [20]: data = [np.random.normal(0, std, 100) for std in range(1, 4)]
plt.violinplot(data, showmedians=True)
plt.xlabel('Distribution')
plt.ylabel('Value')
plt.title('Violin Plot')
plt.xticks(np.arange(1, len(data) + 1), ['Data 1', 'Data 2', 'Data 3'])
plt.legend(['Median'])
plt.show()
```



SEABORN

```
In [48]: import seaborn as sns
```



```
In [49]: data = pd.read_csv(r"C:\Users\surya\OneDrive\Desktop\Project\tips.csv")
data
```

```
Out[49]:
```

	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4
...	...	...	...	...	...	...	...
232	29.03	5.92	Male	No	Sat	Dinner	3
233	27.18	2.00	Female	Yes	Sat	Dinner	2
234	22.67	2.00	Male	Yes	Sat	Dinner	2
235	17.82	1.75	Male	No	Sat	Dinner	2
236	18.78	3.00	Female	No	Thur	Dinner	2

237 rows × 7 columns

```
In [50]: #data understanding
```

```
In [51]: data.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 237 entries, 0 to 236
Data columns (total 7 columns):
 #   Column        Non-Null Count  Dtype  
---  -
 0   total_bill    237 non-null    float64
 1   tip           237 non-null    float64
 2   sex           237 non-null    object  
 3   smoker        237 non-null    object  
 4   day           237 non-null    object  
 5   time          237 non-null    object  
 6   size          237 non-null    int64   
dtypes: float64(2), int64(1), object(4)
memory usage: 13.1+ KB
```

```
In [52]: data.describe()
```

```
Out[52]:
```

	total_bill	tip	size
count	237.000000	237.000000	237.000000
mean	18.984937	2.910759	2.531646
std	7.678298	1.228049	0.922902
min	3.070000	1.000000	1.000000
25%	13.270000	2.000000	2.000000
50%	17.470000	2.750000	2.000000
75%	23.330000	3.500000	3.000000
max	41.190000	7.580000	6.000000

```
In [53]: data.isnull().sum()
```

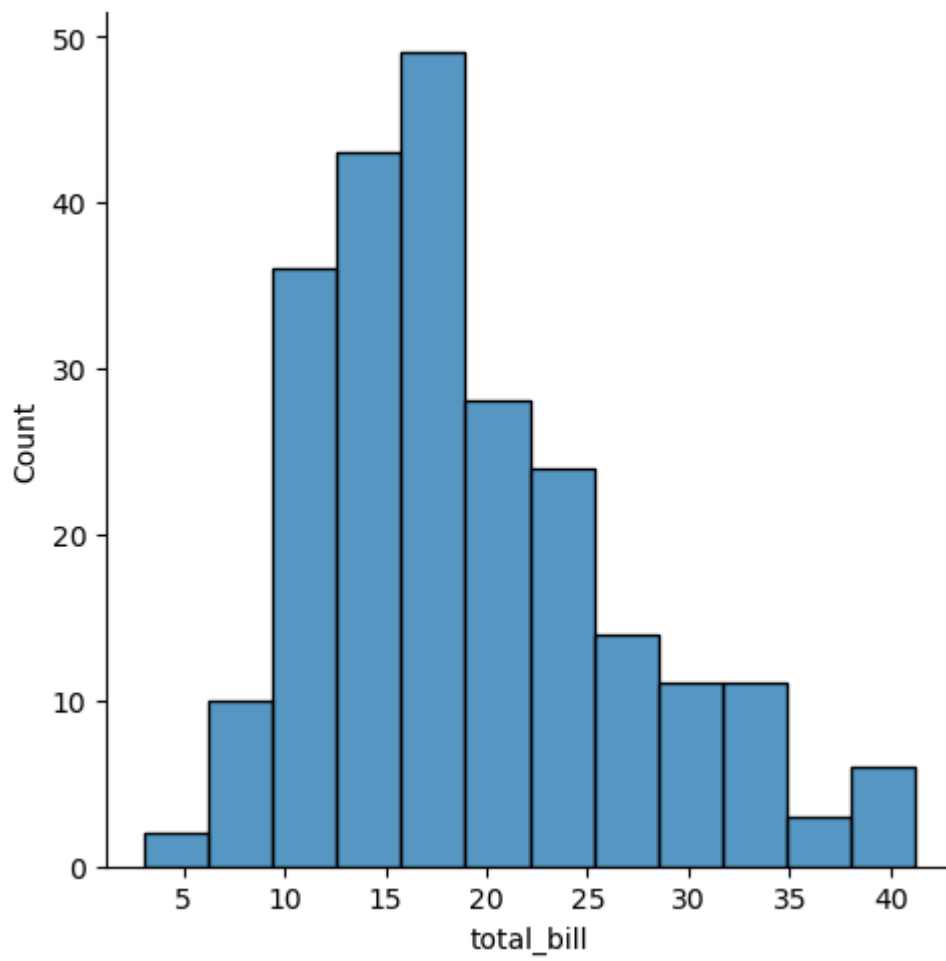
```
Out[53]: total_bill    0
tip                0
sex                0
smoker            0
day               0
time              0
size              0
dtype: int64
```

```
In [54]: data['sex'].value_counts()
```

```
Out[54]: Male        152
Female         85
Name: sex, dtype: int64
```

```
In [55]: sns.displot(data['total_bill'])
```

```
Out[55]: <seaborn.axisgrid.FacetGrid at 0x26c18276908>
```



```
In [56]: sns.distplot(data['total_bill'])
```

```
c:\users\surya\appdata\local\programs\python\python37\lib\site-packages\ipykernel_launcher.py:1: UserWarning:
```

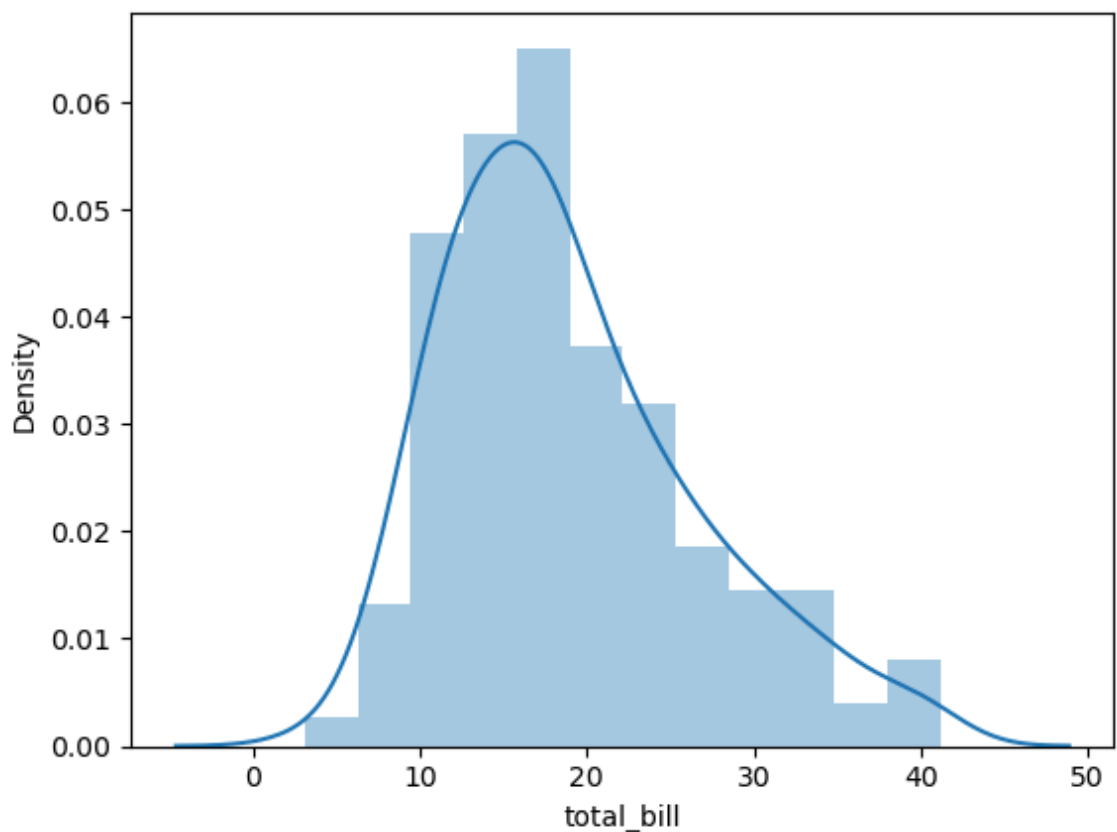
```
`distplot` is a deprecated function and will be removed in seaborn v0.14.0.
```

Please adapt your code to use either `displot` (a figure-level function with similar flexibility) or `histplot` (an axes-level function for histograms).

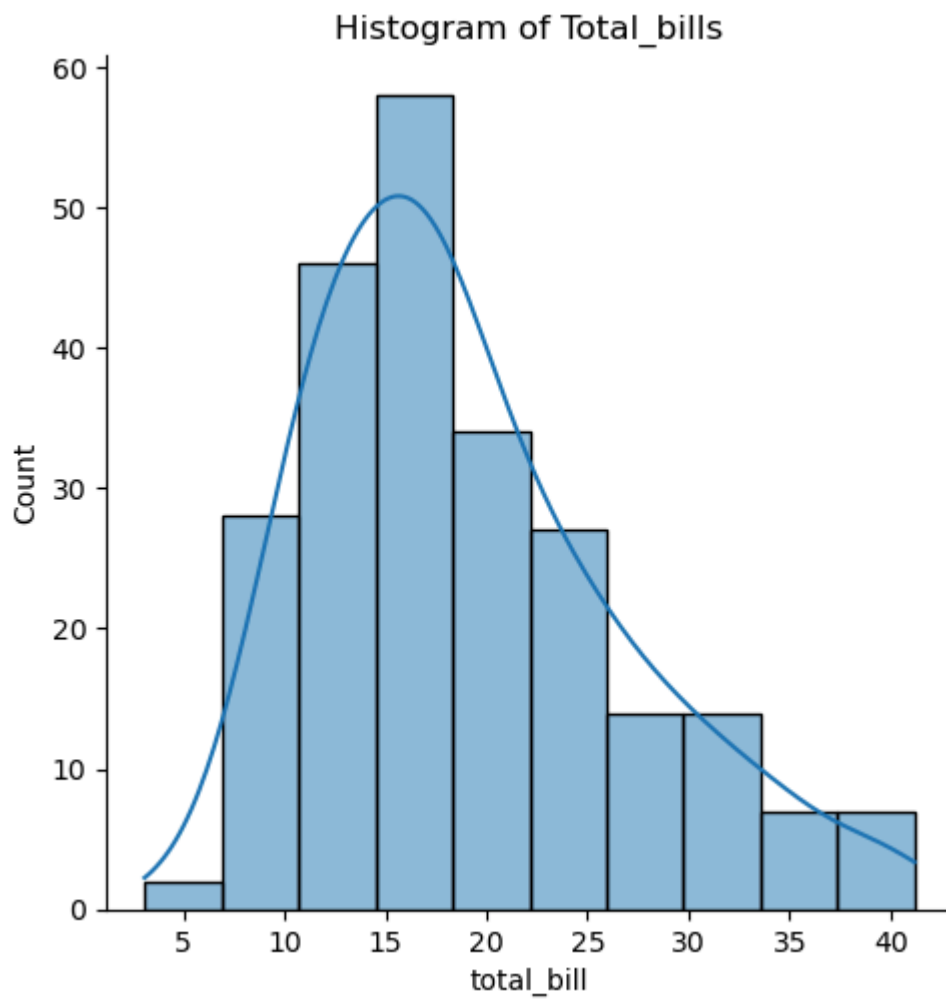
For a guide to updating your code to use the new functions, please see <https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751> (<https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751>)

```
"""Entry point for launching an IPython kernel.
```

```
Out[56]: <matplotlib.axes._subplots.AxesSubplot at 0x26c18182088>
```

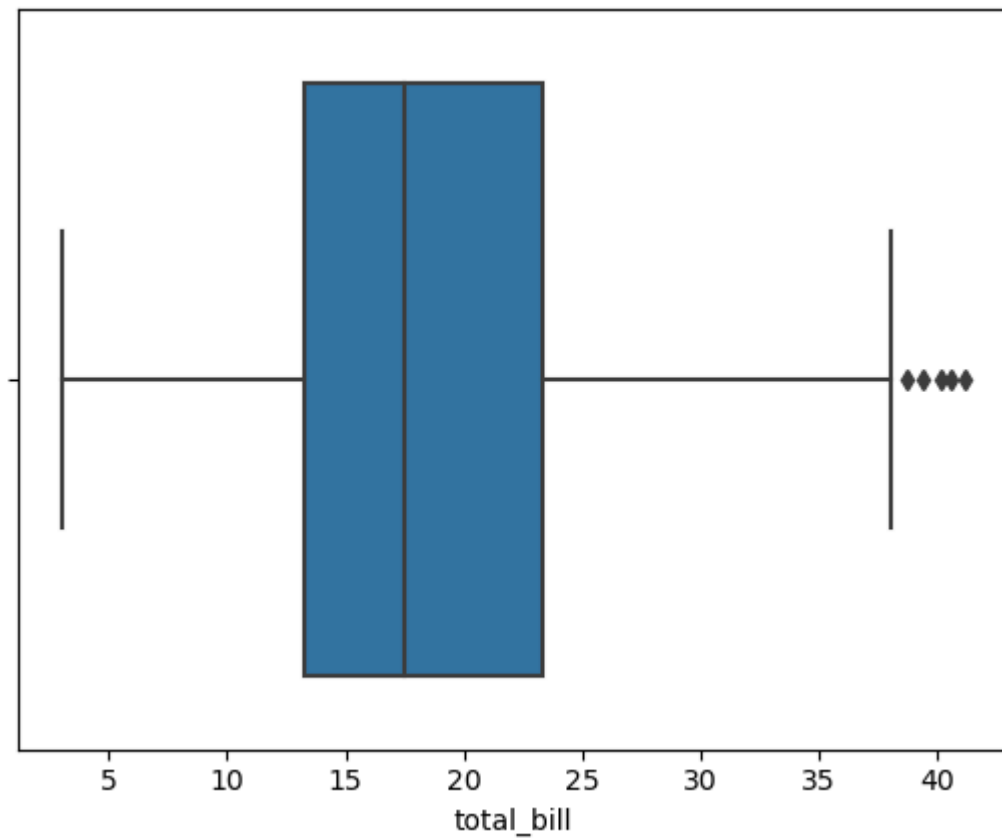


```
In [57]: sns.displot(data['total_bill'],bins=10,kde=True)  
plt.title('Histogram of Total_bills')  
plt.show()
```

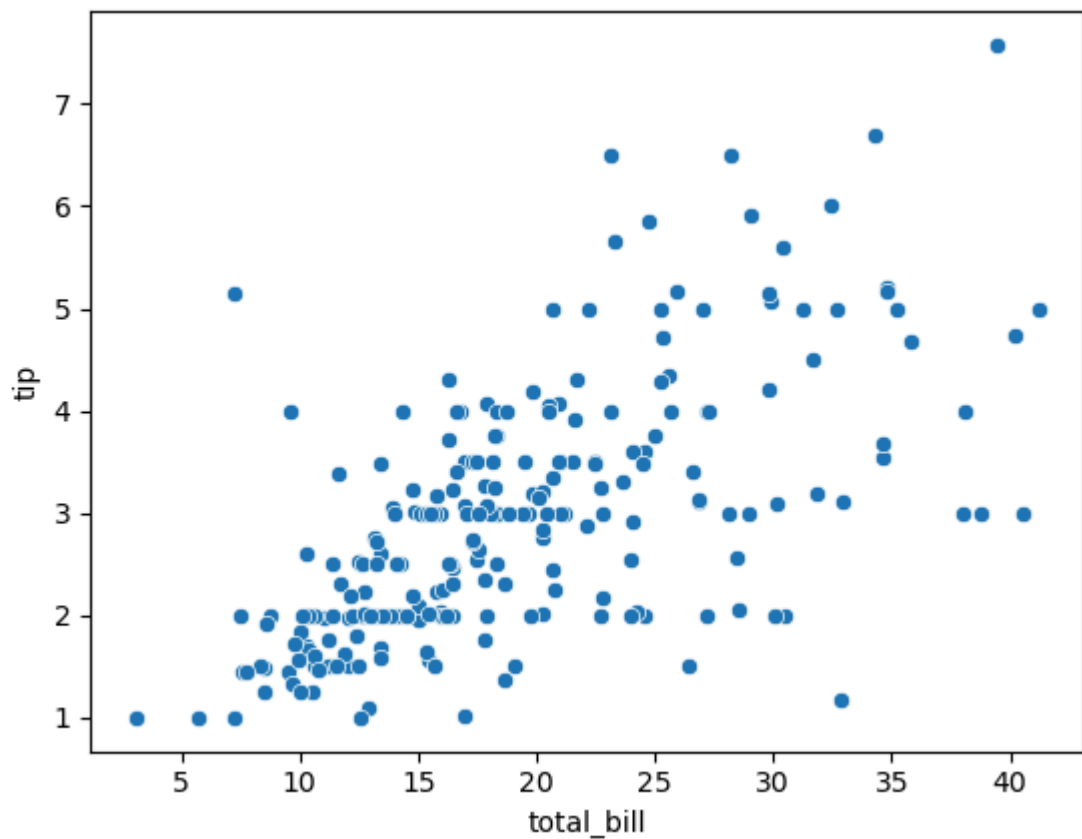


```
In [58]: sns.boxplot(x=data['total_bill'])
```

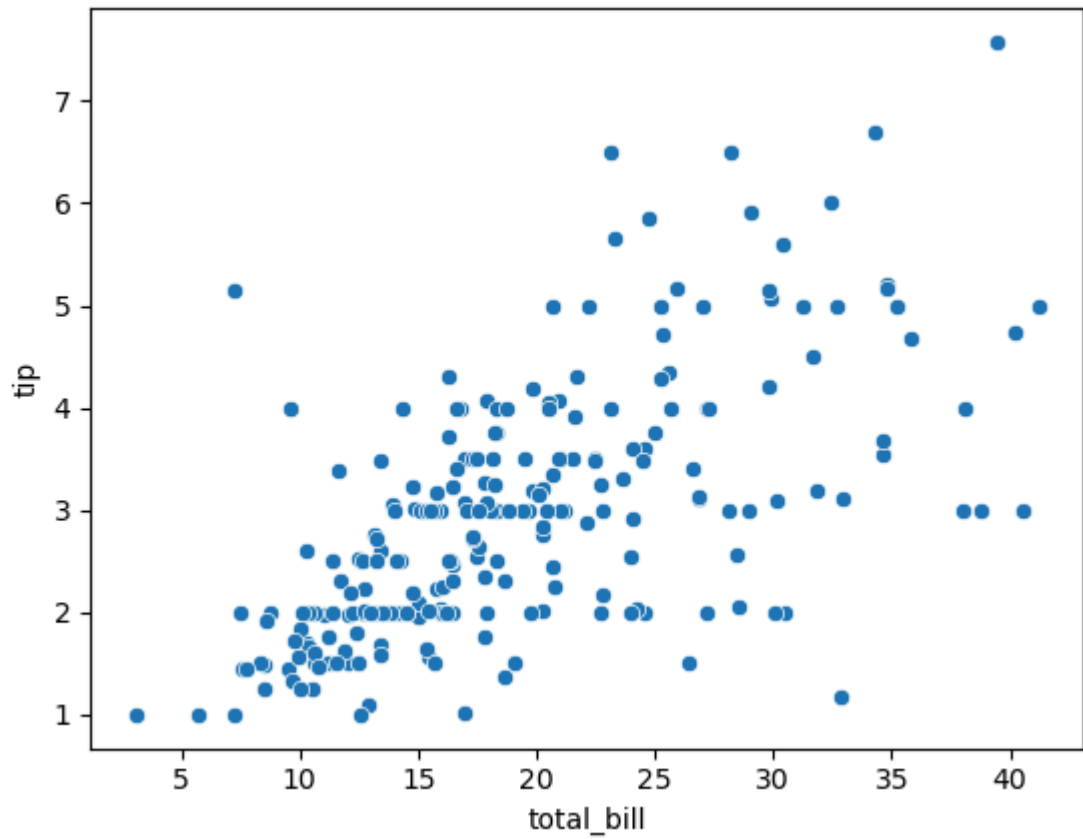
```
Out[58]: <matplotlib.axes._subplots.AxesSubplot at 0x26c18252e48>
```



```
In [59]: sns.scatterplot(x=data['total_bill'],y=data['tip'])  
plt.show()
```

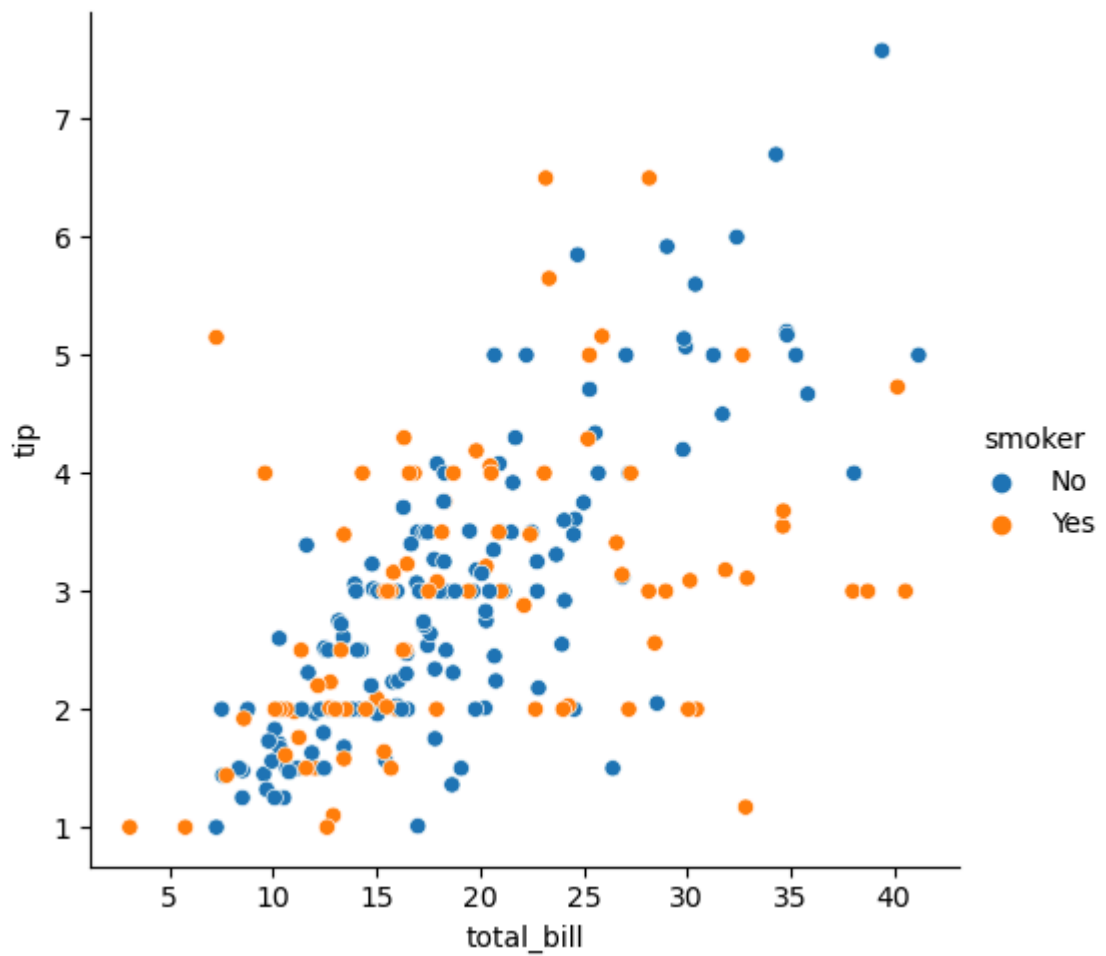


```
In [61]: sns.scatterplot(x=data['total_bill'],y=data['tip'],data=data)  
plt.show()
```



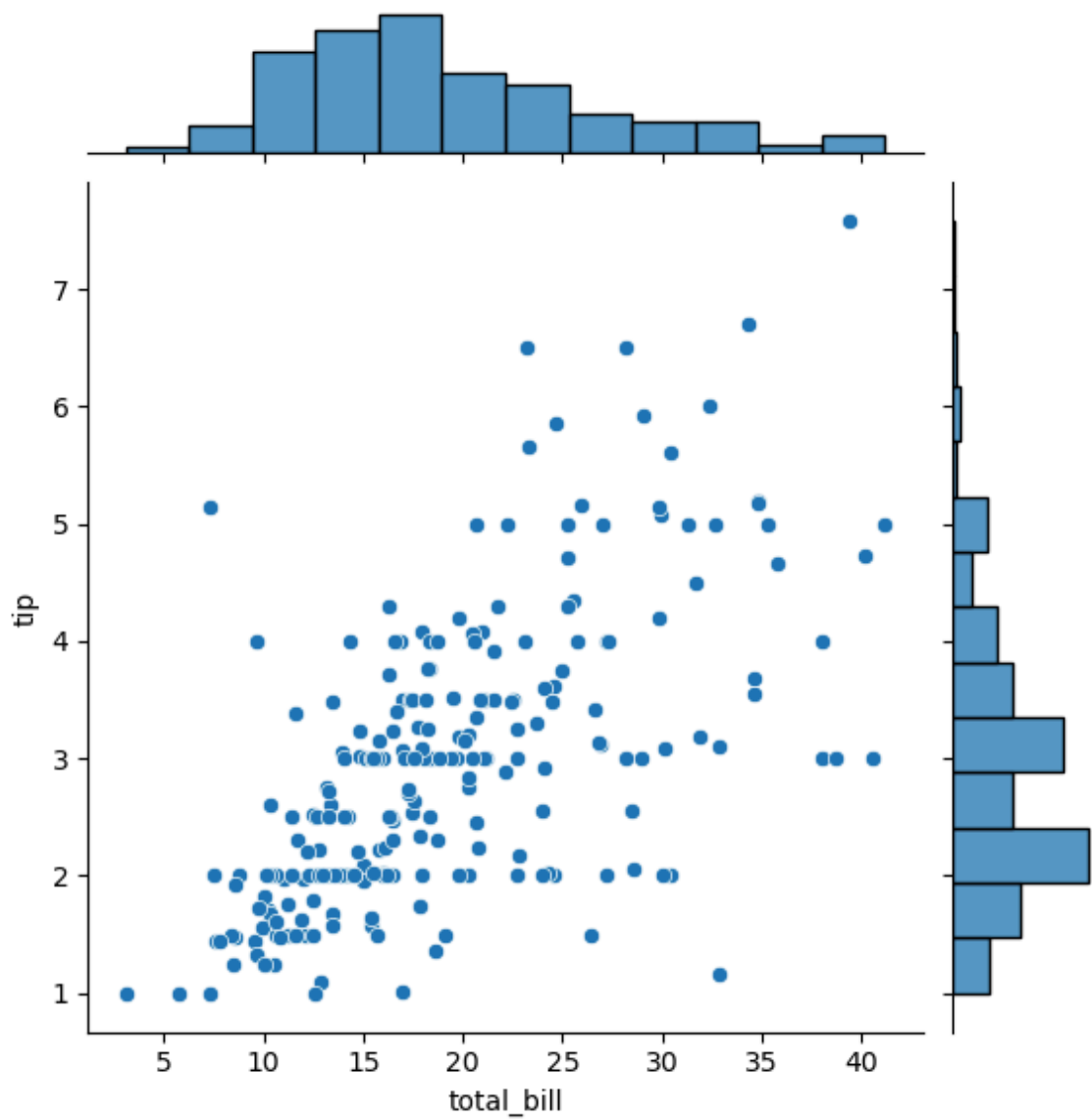
```
In [64]: sns.relplot(x='total_bill',y='tip',data = data,hue='smoker')
```

```
Out[64]: <seaborn.axisgrid.FacetGrid at 0x26c15eeb0c8>
```



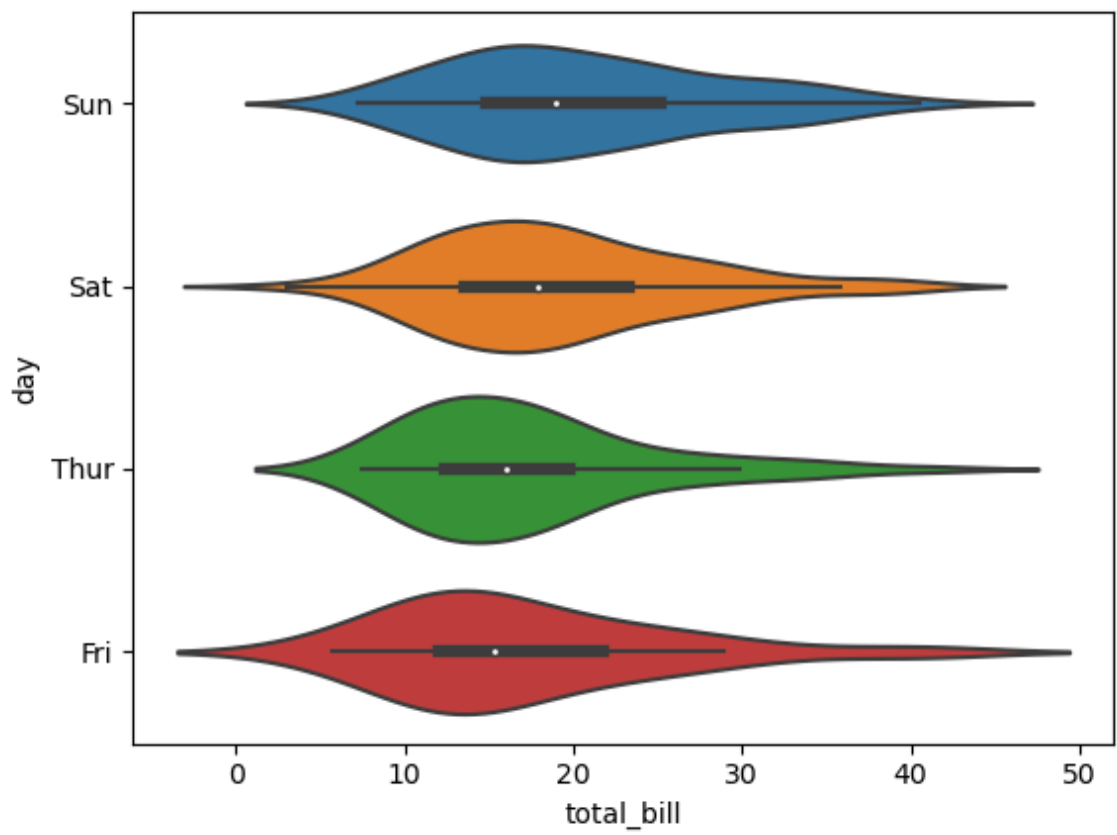

```
In [67]: sns.jointplot(x='total_bill',y='tip',data=data)
```

```
Out[67]: <seaborn.axisgrid.JointGrid at 0x26c1808b148>
```

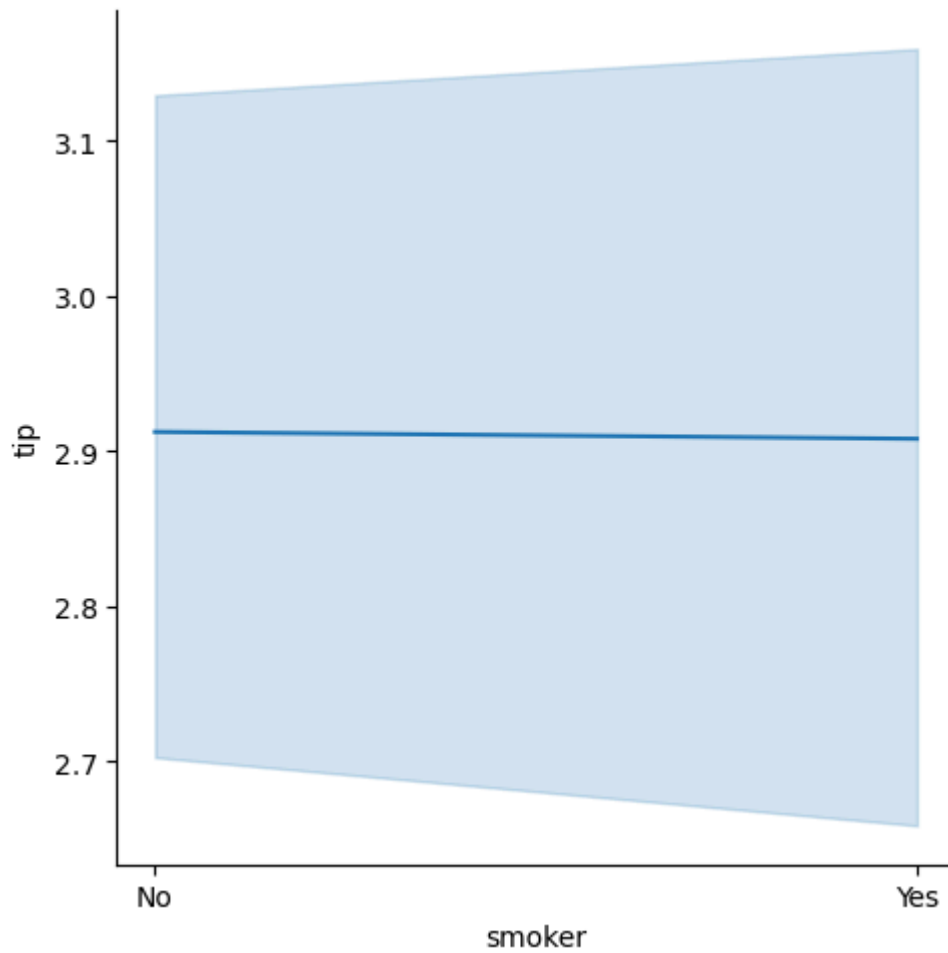


```
In [66]: sns.violinplot(x='total_bill',y='day',data=data)
```

```
Out[66]: <matplotlib.axes._subplots.AxesSubplot at 0x26c18385ac8>
```



```
In [65]: sns.relplot(x='smoker',y='tip',kind='line',data=data)  
plt.show()
```



In []:

In []:

In []:

In []: